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United States Patent	12413857
Kind Code	B2
Date of Patent	September 09, 2025
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Focus-of-expansion locating device

Abstract

The focus-of-expansion locating device includes a loaded state determination unit and a focus-of-expansion locating unit. The loaded state determination unit determines whether the loaded state of the vehicle has changed based on the detection result from the in-vehicle sensor that detects the loaded state of the vehicle. The focus-of-expansion locating unit locates a focus of expansion on the captured image based on the captured image of the camera of the vehicle. The focus-of-expansion locating unit re-locates the focus of expansion when the loaded state has changed.

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Appl. No.:	18/457007
Filed:	August 28, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240179407 A1	May. 30, 2024

Foreign Application Priority Data

JP	2022-187208	Nov. 24, 2022
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Publication Classification

Int. Cl.: H04N23/67 (20230101); G06T7/70 (20170101); G06V20/56 (20220101)

U.S. Cl.:

CPC **H04N23/675** (20230101); **G06T7/70** (20170101); **G06V20/56** (20220101);
G06T2207/10148 (20130101); G06T2207/30242 (20130101); G06T2207/30252
(20130101)

Field of Classification Search

CPC: H04N (23/675); G06T (7/70); G06T (2207/10148); G06T (2207/30242); G06T
(2207/30252); G06V (20/56)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Japanese Patent Application No. 2022-187208 filed on Nov. 24, 2022, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to focus-of-expansion locating devices.

2. Description of Related Art

(3) Japanese Unexamined Patent Application Publication No. 2012-045706 (JP 2012-045706 A) is known as a technical document about a focus-of-expansion locating device. JP 2012-045706 A discloses a technique of locating, based on an image captured by a camera, a focus of expansion in the captured image, and using the focus of expansion to calculate a physical quantity such as time of contact (TOC).

SUMMARY

(4) In the related art as described above, once a focus of expansion is located, this focus of expansion is continuously used for a predetermined period of time. However, if, for example, the orientation of the camera changes during the predetermined period of time, the position of the focus of expansion in the captured image may deviate from a correct position.

(5) It is an object of the present disclosure to provide a focus-of-expansion locating device that can re-locate a focus of expansion at an appropriate timing.

(6) A focus-of-expansion locating device of the present disclosure includes: a loaded state determination unit configured to determine whether a loaded state of a vehicle has changed based on a detection result from an in-vehicle sensor configured to detect the loaded state of the vehicle; and a focus-of-expansion locating unit configured to locate, based on an image captured by a camera of the vehicle, a focus of expansion in the captured image. The focus-of-expansion locating unit re-locates the focus of expansion when the loaded state has changed.

(7) According to the present disclosure, a focus-of-expansion locating device can be provided that can re-locate a focus of expansion at an appropriate timing.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

(2) FIG. 1 is a block diagram of a focus-of-expansion locating device according to an embodiment;

(3) FIG. 2A is a diagram illustrating a change in attitude of the vehicle when the loaded state has changed;

(4) FIG. 2B is a diagram illustrating a change in attitude of the vehicle when the loaded state has changed;

(5) FIG. 3A is a diagram illustrating a change in focus of expansion when the loaded state has changed;

(6) FIG. 3B is a diagram showing a change in focus of expansion when the loaded state has changed;

(7) FIG. 4 is a diagram illustrating a search area for locating a focus of expansion;

(8) FIG. 5 is a diagram illustrating a change in focus of expansion when the loaded state has changed; and

(9) FIG. 6 is a flowchart showing a process performed by the focus-of-expansion locating device shown in FIG. 1.

DETAILED DESCRIPTION OF EMBODIMENTS

(10) Hereinafter, embodiments of the present disclosure will be described with reference to the drawings.

(11) FIG. 1 is a block diagram of a focus-of-expansion locating device according to an embodiment. As illustrated in FIG. 1, the focus-of-expansion locating device 1 includes a camera 2, an in-vehicle sensor 3, a display 4, and an Electronic Control Unit (ECU) 10. The focus-of-

expansion locating device **1** is a device for locating a focus of expansion (FOE).

(12) The focus of expansion is a point of infinite distance. A plurality of straight lines parallel to the predetermined direction intersects at focuses of expansion when viewed from the predetermined direction. When the viewpoint (e.g., the human eye) is advancing along a direction parallel to the line of sight, a plurality of objects appearing in the field of view move along a radial trajectory starting from the focus of expansion.

(13) The camera **2** is provided in the cabin of the vehicle **20** (see FIG. 2A and FIG. 2B). The camera **2** is fixed to the vehicle **20**. The camera **2** captures an image of the front of the vehicle **20** via, for example, a windshield of the vehicle **20**. The camera **2** is, for example, a monocular camera. The camera **2** may be, for example, a stereo camera. The camera **2** transmits the captured images to ECU **10**.

(14) The in-vehicle sensor **3** detects a loaded state of the vehicle **20**. The loaded state of the vehicle **20** is an index corresponding to a weight of a load, an occupant, or the like conveyed by the vehicle **20**. In the present embodiment, the loaded state of the vehicle **20** is the number of towed vehicles **30** (see FIG. 2B of the drawings) connected to the vehicle **20**. That is, the in-vehicle sensor **3** detects the number of towed vehicles **30** connected to the vehicle **20**. The in-vehicle sensor **3** is, for example, a Trailer detection module (TDM). The in-vehicle sensor **3** sends the loaded state of the vehicles **20** to the ECU **10**.

(15) The display **4** is provided, for example, on an instrument panel of the vehicle **20**. The display **4** has a display screen for displaying an image. The display **4** displays images on a display screen based on control signals from ECU **10**. The display **4** displays, for example, a captured image of the camera **2**. The display **4** may be, for example, a windshield of the vehicle **20** or a Head Up Display (HUD) for displaying a projection on a projection screen.

(16) ECU **10** is an electronic control unit having a Central Processing Unit (CPU) and a storage. The storage unit is, for example, Read Only Memory (ROM) or Random Access Memory (RAM). In ECU **10**, for example, various functions are realized by executing a program stored in a storage unit by a CPU.

(17) FIG. 2A and FIG. 2B are diagrams illustrating a change in attitude of the vehicle **20** when the loaded state has changed. FIG. 2A illustrates a case where the loaded state of the vehicles **20** is the first state. FIG. 2B illustrates a case where the loaded state of the vehicles **20** is the second state. The number of towed vehicles **30** in the second state is larger than the number of towed vehicles **30** in the first state. In the present embodiment, the number of towed vehicles **30** in the first state is "0", and the number of towed vehicles **30** in the second state is "1".

(18) As shown in FIG. 2A, in the first condition, the optical axis **21** of the camera **2** is substantially parallel to the traveling direction (e.g., horizontal direction) D of the vehicle **20**. That is, in the first state, the camera **2** faces the front of the vehicle **20** in the horizontal direction. As shown in FIG. 2B, in the second state, since the towed vehicle **30** is connected to the rear of the vehicle **20**, the front end of the vehicle **20** is displaced upward in the vertical direction and the rear end of the vehicle **20** is displaced downward in the vertical direction as compared with the first state. That is, the attitude of the vehicle **20** in the first state and the attitude of the vehicle **20** in the second state are different from each other.

(19) Since the camera **2** is fixed to the vehicle **20**, the orientation of the camera **2** also changes when the attitude of the vehicle **20** changes. In the second state, the optical axis **21** of the camera **2** intersects the traveling direction D of the vehicle **20**. That is, in the case of the second state, the camera **2** faces upward in the vertical direction with respect to the horizontal direction. As described above, even when the traveling direction D of the vehicle **20** has not changed, the orientation of the camera **2** may change. When the orientation of the camera **2** changes, the position of the focus of expansion in the captured image of the camera **2** also changes. Therefore, for example, if the focus of expansion of the first state continues to be used even in the second state, it may be difficult to accurately calculate the physical quantity of the second state.

(20) ECU **10** includes, as a functional configuration, a loaded state determination unit **11**, a focus-of-expansion locating unit **12**, a focus-of-expansion storage unit **13**, and a physical quantity calculation unit **14**.

(21) The loaded state determination unit **11** determines the loaded state of the vehicle **20** based on the detection result of the in-vehicle sensor **3**. The loaded state determination unit **11** determines whether or not the towed vehicle **30** is connected to the vehicle **20**. The loaded state determination unit **11** determines the number of towed vehicles **30** connected to the vehicle **20**.

(22) The loaded state determination unit **11** determines whether or not the loaded state of the vehicle **20** has changed based on the detection result of the in-vehicle sensor **3**. The loaded state determination unit **11** determines whether or not the number of towed vehicles **30** connected to the vehicle **20** has increased. The loaded state determination unit **11** determines whether or not the number of towed vehicles **30** connected to the vehicle **20** has decreased. The loaded state determination unit **11** determines that the loaded state of the vehicle **20** has changed when the number of towed vehicles **30** connected to the vehicle **20** has changed.

(23) The focus-of-expansion locating unit **12** locates a focus of expansion on the captured image based on the captured image of the camera **2** of the vehicle **20**. The focus-of-expansion locating unit **12** re-locates a focus of expansion when the loaded state of the vehicle **20** has changed. The focus-of-expansion locating unit **12** locates a focus of expansion each time the loaded state of the vehicle **20** changes. The focus-of-expansion locating unit **12** locates a focus of expansion by various known methods.

(24) FIG. 3A shows captured images of the camera **2** in the first condition. As shown in FIG. 3A, the captured images **22** have, for example, a pair of lane boundary lines **40** that are parallel to each other. The focus-of-expansion locating unit **12** extracts the pair of lane boundary lines **40** and locates the intersection of the extension lines of the pair of lane boundary lines **40** as the focus of expansion **51** in the first state.

(25) The abscissa X1 of the focus of expansion **51** in the captured image **22** is, for example, 100 pixels. The ordinate Y1 of the focus of expansion **51** in the captured image **22** is, for example, 100 pixels. That is, the focus of expansion **51** is 100 pixels away from the left end of the captured image **22** and 100 pixels away from the lower end of the captured image **22**.

(26) FIG. 3B shows the captured images of the cameras **2** in the second condition. The focus-of-expansion locating unit **12** extracts the pair of lane boundary lines **40** and locates the intersection of the extension lines of the pair of lane boundary lines **40** as the focus of expansion **52** in the second state. As described above, in the case of the first state, the camera **2** faces the front of the vehicle **20** in the horizontal direction, whereas in the case of the second state, the camera **2** faces the upper side in the vertical direction with respect to the horizontal direction. Therefore, in the captured image **22**, the focus of expansion **52** in the second state is located below the focus of expansion **51** in the first state.

(27) The abscissa X2 of the focus of expansion **52** in the captured image **22** is the same as the abscissa X1 of the focus of expansion **51** in the captured image **22**. The ordinate Y2 of the focus of expansion **52** in the captured image **22** is smaller than the ordinate Y1 of the focus of expansion **51** in the captured image **22**. The ordinate Y2 of the focus of expansion **52** in the captured image **22** is, for example, 80 pixels. That is, the focus of expansion **52** is 100 pixels away from the left end of the captured image **22** and 80 pixels away from the lower end of the captured image **22**.

(28) The focus-of-expansion storage unit **13** stores the focuses of expansion **51**, **52** located by the focus-of-expansion locating unit **12**. The focus-of-expansion storage unit **13** stores the focus of expansion **51** in the case where the loaded state of the vehicle **20** is the first state as the first focus of expansion, and stores the focus of expansion **52** in the case where the loaded state of the vehicle **20** is the second state as the second focus of expansion. The focus-of-expansion storage unit **13** stores focuses of expansion for each loaded state of the vehicle **20**.

(29) The focus-of-expansion locating unit **12** locates a focus of expansion in a predetermined

search area of the captured image **22** using the initial position as a starting point. FIG. **4** is a diagram illustrating a search area for locating a focus of expansion. As illustrated in FIG. **4**, in the case of the first state, the focus-of-expansion locating unit **12** locates the focus of expansion **51** in the search area **61** using the initial position as a starting point. The initial position is the position of the focus of expansion immediately before the change in loaded state. When calculating the first focus of expansion after the start of the vehicle **20**, the focus-of-expansion locating unit **12** may locate a focus of expansion using a pre-stored initial position as a starting point.

(30) When the loaded state of the vehicle **20** has changed, the focus-of-expansion locating unit **12** re-locates a new focus of expansion after the change in loaded state by using the position of the focus of expansion immediately before the change in loaded state as an initial position. When the vehicle **20** has changed from the first state to the second state, the focus-of-expansion locating unit **12** re-locates a focus of expansion **52** in the search area **62** using the position of the focus of expansion **51** (first focus of expansion) as an initial position. The search area **62** is larger than the search area **61**. That is, when the loaded state of the vehicle **20** has changed, the focus-of-expansion locating unit **12** expands the search area and re-locates a new focus of expansion.

(31) When the focus of expansion **51** is stored in the focus-of-expansion storage unit **13** and the loaded state of the vehicle **20** has changed from the other than the first state to the first state, the focus-of-expansion locating unit **12** re-locates a focus of expansion using the position of the focus of expansion **51** as an initial position. For example, when the loaded state of the vehicle **20** changed from the first state to the second state and then has returned to the first state, the focus-of-expansion locating unit **12** re-locates the focus of expansion when the loaded state of the vehicle **20** has returned to the first state, by using the position of the focus of expansion **51** (first focus of expansion) rather than the focus of expansion **52** (second focus of expansion) as an initial position.

(32) FIG. **5** is a diagram illustrating a change in focus of expansion when the loaded state of the vehicle **20** has returned to the first state after changing from the first state to the second state. As shown in FIG. **5**, the position of the focus of expansion **53** when the loaded state of the vehicle **20** is returned to the first state after changing from the first state to the second state may be different from the position of the focus of expansion **51** of the first state before changing to the second state. This is considered to be due to, for example, a decrease in the fuel of the vehicle **20** or a change in load weight of the towed vehicle **30** while the vehicle **20** returns to the first state after the loaded state changed from the first state to the second state.

(33) The abscissa X3 of the focus of expansion **53** in the captured image **22** is the same as the abscissa X1 of the focus of expansion **51** in the captured image **22**. The ordinate Y3 of the focus of expansion **53** in the captured image **22** is larger than the ordinate Y1 of the focus of expansion **51** in the captured image **22**. The ordinate Y3 of the focus of expansion **53** in the captured image **22** is, for example, 120 pixels. That is, the focus of expansion **53** is 100 pixels away from the left end of the captured image **22** and 120 pixels away from the lower end of the captured image **22**.

(34) Although the position of the focus of expansion **53** is different from the position of the focus of expansion **51**, since the focus of expansion **53** and the focus of expansion **51** are focuses of expansion in the same loaded state (first state), the position of the focus of expansion **53** tends to be closer to the focus of expansion **51** than the focus of expansion **52** in the second state. Therefore, in a case where the focus of expansion **53** is located using the position of the focus of expansion **51** as an initial position, the learning efficiency of the focus of expansion is improved as compared with the case where the focus of expansion **53** is located using the position of the focus of expansion **52** as an initial position.

(35) The physical quantity calculation unit **14** calculates each physical quantity based on the latest focus of expansion located by the focus-of-expansion locating unit **12**. The physical quantity calculation unit **14** calculates the distance between the obstacle and the vehicle **20** based on, for example, the distance between the obstacle and the focus of expansion in the captured image **22**.

(36) Next, the ECU **10** process will be described. FIG. **6** is a flow chart illustrating a process

performed by the ECU **10**. As shown in FIG. **6**, the ECU **10** locates the focus of expansion **51** in the first condition in **S1**.

(37) The ECU **10** determines, in **S2**, whether the loaded state of the vehicles **20** has changed from the first state to the second state. That is, the ECU **10** determines whether or not the number of towed vehicles **30** connected to the vehicle **20** has increased to the number of second-state vehicles in **S2**. When the loaded state of the vehicles **20** has changed from the first state to the second state (**S2**: YES), the ECU **10** proceeds to **S3**. When the loaded state of the vehicles **20** does not change from the first state to the second state (**S2**: NO), the ECU **10** ends the present process.

(38) The ECU **10** stores, in **S3**, the focus of expansion **51** in the first condition as the first focus of expansion. The ECU **10** expands the search area in **S4**. The ECU **10** locates a focus of expansion **52** in the second condition in **S5**. In **S5**, the ECU **10** locates the focus of expansion **52** in the second state in the search area **62** enlarged by **S4** with the position of the focus of expansion **51** in the first state as the starting position.

(39) The ECU **10** determines, in **S6**, whether the loaded state of the vehicles **20** has changed from the second state to the first state. That is, the ECU **10** determines, in **S6**, whether or not the number of towed vehicles **30** connected to the vehicle **20** has decreased to the first number. When the loaded state of the vehicles **20** has changed from the second state to the first state (**S6**: YES), the ECU **10** proceeds to **S7**. When the loaded state of the vehicles **20** does not change from the second state to the first state (**S6**: NO), the ECU **10** ends the present process.

(40) In **S7**, the ECU **10** sets the position of the focus of expansion **51** (the first focus of expansion) in the first state stored in **S3** as the default position, instead of the focus of expansion **52** (the second focus of expansion) in the second state. In **S8**, the ECU **10** locates a new focus of expansion **53** in the first condition starting from the default position set in **S7**.

(41) As described above, in the focus-of-expansion locating device **1**, the focus-of-expansion locating unit **12** re-locates the focus of expansion when the loaded state of the vehicle **20** has changed. Thus, even if the orientation of the camera **2** has changed due to a change in loaded state of the vehicle **20**, it is possible to re-locate a new focus of expansion at an appropriate timing. That is, since the focus-of-expansion locating unit **12** locates a focus of expansion by using a change in loaded state of the vehicle **20** as a trigger, it is possible to re-locate a new focus of expansion without waiting until the normal location start condition is satisfied. Therefore, it is possible to prevent the focus of expansion deviated from the accurate position from being used for calculating the physical quantity. Therefore, according to the focus-of-expansion locating device **1**, it is possible to calculate the physical quantity with high accuracy.

(42) The in-vehicle sensor **3** detects the number of towed vehicles **30** connected to the vehicle **20**. The loaded state determination unit **11** determines that the loaded state of the vehicle **20** has changed when the number of towed vehicles **30** has changed. When the number of towed vehicles **30** changes, a change in attitude of the vehicle **20** tends to be relatively large. When the change in attitude of the vehicle **20** is large, the change in orientation of the camera **2** of the vehicle **20** also tends to be relatively large. In such a case, there is a possibility that the focus of expansion may deviate from the correct position by a relatively large amount, and the above-described effect becomes particularly remarkable.

(43) The focus-of-expansion storage unit **13** stores the focus of expansion **51** in a case where the loaded state of the vehicle **20** is the first state as the first focus of expansion. In the case where the first focus of expansion is stored in the focus-of-expansion storage unit **13**, the focus-of-expansion locating unit **12** re-locates the focus of expansion **52** using the position of the focus of expansion **51** as an initial position when the loaded state of the vehicle **20** has changed from a state other than the first state to the first state. Accordingly, when the loaded state of the vehicle **20** has returned from a state other than the first state to the first state, the position of the focus of expansion **51** (first focus of expansion) is used as an initial position to re-locate the new focus of expansion **53** in the first state, and therefore, it is possible to locate the new focus of expansion **53** in the first state more

efficiently than in the case where the position of the focus of expansion 52 (second focus of expansion) is used as the initial position to re-locate the new focus of expansion 53 in the first state.

(44) Although the embodiments of the present disclosure have been described above, the present disclosure is not limited to the above-described embodiments. The present disclosure may be embodied in various forms with various changes and modifications, including the above-described embodiments, based on the knowledge of those skilled in the art.

(45) Although an example has been shown in which the camera 2 of the vehicle 20 captures an image of the front of the vehicle 20, the camera 2 may capture an image of the rear of the vehicle 20 via the rear glass of the vehicle 20.

(46) Although an example has been shown in which the number of towed vehicles 30 in the first state is “0” and the number of towed vehicles 30 in the second state is “1”, the number of towed vehicles 30 in the first state may be “1” or more and the number of towed vehicles 30 in the second state may be “2” or more. The number of towed vehicles 30 in the first state may be different from the number of towed vehicles 30 in the second state. The number of towed vehicles 30 in the second state may be smaller than the number of towed vehicles 30 in the first state.

(47) Although an example of the number of towed vehicles 30 is shown as the loaded state of the vehicle 20, the loaded state of the vehicle 20 may be an index corresponding to the weight of the occupant. In this case, the in-vehicle sensor 3 may detect, for example, the number of seatbelts mounted on the vehicle 20. The loaded state determination unit 11 may determine that the loaded state of the vehicle 20 has changed when the number of seatbelts attached has changed. The in-vehicle sensor 3 may be a load sensor provided in a seat or a load sensor provided in a load bed.

(48) ECU 10 may not include the physical quantity calculation unit 14. That is, the focuses of expansion 51, 52, and 53 located by the focus-of-expansion locating unit 12 may not be used to calculate the physical quantity. The vehicle 20 may be an autonomous vehicle.

Claims

1. A focus-of-expansion locating device comprising: a processor; and a memory storing instructions that are executable by the processor and that cause the processor to determine whether a loaded state of a vehicle has changed based on a detection result from an in-vehicle sensor configured to detect the loaded state of the vehicle, locate, based on an image captured by a camera of the vehicle, a focus of expansion in the captured image, and re-locate the focus of expansion when the loaded state has changed, wherein the in-vehicle sensor is configured to detect the number of towed vehicles connected to the vehicle, and the processor is configured to determine that the loaded state has changed when the number of towed vehicles has changed.
 2. The focus-of-expansion locating device according to claim 1, wherein the processor is configured to: store the focus of expansion when the loaded state is a first state as a first focus of expansion; and re-locate the focus of expansion using a position of the first focus of expansion as an initial position when the first focus of expansion is stored and the loaded state has changed from a state other than the first state to the first state.
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